Mr. Honghao Fu

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[Homepage] [Github] [Google Scholar]

Education background

The University of Queensland (UQ) **Brisbane, Australia**
- *Doctor of Philosophy in Computer Science* **07/2025-Now**
- Research Interests: Large Language Models, Agentic Systems, Multimodal Understanding
- Supervisor: Prof. Yujun Cai, Prof. Miao Xu and Prof. Yiwei Wang

Nanyang Technological University (NTU) **Singapore**
- *Master of Science in Electronics* **08/2023-01/2025**
- Research Interests: Generative AI, Computer Vision
- Supervisor: Prof. Bihan Wen and Dr. Yufei Wang

Southeast University (SEU) **Nanjing, China**
- *Bachelor of Engineering in Measuring Control Technology & Instruments* **08/2019-06/2023**
- Outstanding Dissertation Award (Top 1%)

Publications

1. **Fu H.**, Xu M., Wang Y., Zhang D., Liu J. & Cai Y. (2026). VideoStir: Understanding Long Videos via Spatio-Temporally Structured and Intent-Aware RAG. *ACL 2026 Main*. (Top 0.8%) [Paper]

2. **Fu H.**, Chen J., Lin M., Zheng J., Kong X., Wang Y., Yu X., Xu M., Jiang Y., & Cai Y. (2026). AdaM-Rec: Adaptive Modality Routing for Multimodal Recommendation. *NeurIPS 2026*. [Paper]

3. **Fu H.**, Ouyang Y., Chang K., Wang Y., Huang Z. & Cai Y. (2026). ContextNav: Towards Agentic Multimodal In-Context Learning. *ICLR 2026*. [Paper]

4. **Fu H.**, Chen J., Mei L., ..., & Cai Y. (2026). From Language to World States: A Survey of World Models in Multimodal Reasoning. Accepted by *Foundations and Trends in Computer Graphics and Vision.* (Q1, IF=18.6)

5. Zhang Z.*, **Fu H.***, Xu M., Wang Y. & Cai Y. (2026). V-DEAL: Diagnosing Video Safety De-Calibration as an Understanding-Refusal Coupling Failure. *COLM 2026*. [Paper]

6. Zhang D., Hu S., **Fu H.**, Feng X., Wang Y., Cheong H. & Huang K. (2026). Rethinking Temporal Modeling in Visual Object Tracking via Decoupled Auxiliary Supervision. *ECCV 2026*. [Paper]

7. Ren J., Zhang G., **Fu H.**, Wu P. & Wang H. (2026). WaMo: Wavelet-Enhanced Multi-Frequency Trajectory Analysis for Fine-Grained Text-Motion Retrieval. *ACM-MM 2026*. [Paper]

8. Yang Y., Cheng W., Chen S., **Fu H.**, Zeng X., Cai Y., Yu G. & Ma X. (2026). OmniLottie: Generating Vector Animations via Parameterized Lottie Tokens. *CVPR 2026*. [Paper]

9. Chen J., Zhang T., Lin M., Huang D., Shi T., **Fu H.**, … , Kong X. & Jiang Y. (2026). ShopX: A Foundation Model for Intent-to-Item Fulfillment in Agentic Shopping. *arXiv: 2606.31693*. (Technical Report) [Paper]

10. **Fu H.***, Ren J.*, Chai Q., Ye D., Cai Y. & Wang H. (2025). VistaWise: Building Cost-Effective Agent with Cross-Modal Knowledge Graph for Minecraft. *EMNLP 2025 Main*. [Paper]

11. **Fu H.**, Shen Z., Chin J. J., & Wang H. (2025). BrainVis: Exploring the Bridge between Brain and Visual Signals via Image Reconstruction. *ICASSP 2025*. [Paper]

12. **Fu H.**, Gu Y., Yan Y., Shen Y., Wu Y. & Sun L. (2025). SDR-GAIN: A High Real-Time Occluded Pedestrian Pose Completion Method for Autonomous Driving. *IEEE Transactions on Intelligent Transportation Systems*. (Q1, IF=8.4) [Paper]

13. Gu Y., Yan X., Qin H., Akhtar N., Yuan S., **Fu H.**, Yang S. & Mian A. (2025).HDTCNet:A hybrid-dimension al convolutional network for multivariate time series classification. *Pattern Recognition*. (Q1, IF=7.5) [Paper]

14. Liu S., Han Y., Xing P., Yin F., Wang R., Cheng W., Liao J., Wang Y., **Fu H.**, … , Yu G. & Jiang D. (2025). Step1X-Edit: A Practical Framework for General Image Editing. *arXiv:2504.17761*. (Technical Report) [Paper]

15. **Fu H.***, Wang Y.*, Yang W., & Wen B. (2024). DP-IQA: Utilizing Diffusion Prior for Blind Image Quality Assessment in the Wild. arXiv preprint *arXiv:2405.19996*. (Under Review) [Paper]

16. **Fu H.***, Shen Y.*, Liu Y., Li J., & Zhang X. (2023). SGCN: A multi-order neighborhood feature fusion landform classification method based on superpixel and graph convolutional network. *International Journal of Applied Earth Observation and Geoinformation*. (Q1, IF=8.6) [Paper]

Patent

1. A Hidden Cable Positioning Detection and AR Visualization System, Patent No. CN114089271A, 2022. [Link]

Work Experience

StepFun **Shanghai, China**
Research Intern **03/2025-06/2025**

- Worked on data curation for generative AI. Trained T2I diffusion models tailored to business applications.
- Participated in the development of the diffusion-based image editing model Step1X-Edit and the VLM-based vector animation generation model LottieGen. Contributed to the corresponding technical report and paper.

Hong Kong University of Science and Technology (Guangzhou) **Remote**
Research Assistant **08/2023-01/2025**

- Worked on LLM-based agents in video games, 2D/3D generative model and scene reconstruction, and multimodal alignment between biological signals and images. Published two research papers.

Sun Yat-sen University **Remote**
Research Assistant **11/2022-06/2023**

- Worked on GNN-based semantic segmentation of remote sensing images and published one research papers.

Competition

SEU Robo Cup Robot Competition: Visual Group, Prize of Excellence **09/2020**
SEU Smart Car Competition: Visual Group, Prize of Third Place **11-12/2019**

Technical Skills

- **Programming skills:** Python (PyTorch), C/C++, HTML, Matlab and vibe coding tools.
- **Software skills:** Linux, Visual Studio, PyCharm and InterlliJ IDEA.

Teaching & Services

- **Teaching:** UQ Summer Research Program Tutor.
- **Service:** Reviewer of ICLR, NeurIPS, ICML, ACL, CVPR, ACM-MM, CIKM, IJCNN, ICASSP, VCIP, IEEE Transactions on Multimedia (T-MM), IEEE Transactions on Intelligent Transportation Systems (T-ITS), IEEE Internet of Things Journal (IoT), Journal of Selected Topics in Signal Processing (J-STSP), Pattern Recognition (PR), Neural Networks, Neurocomputing, Machine Learning with Applications, etc.